

Progress Update #4: Feature Selection Pipeline

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Pipeline Overview

Stage 1: 17 FS methods → each outputs a 0/1 mask



Stage 2: 3 predictors (KNN, SVR, RF) × LOOCV → MAE, Kappa, ...



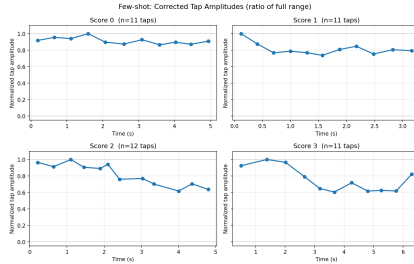
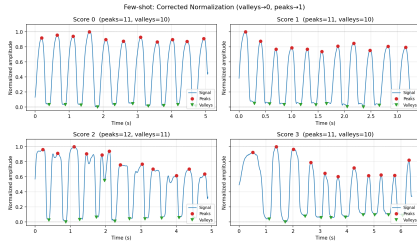
Stage 3: Rank features by inclusion frequency (good FS only)



Stage 4: Feed top-K features to LLM (32B) → find best K



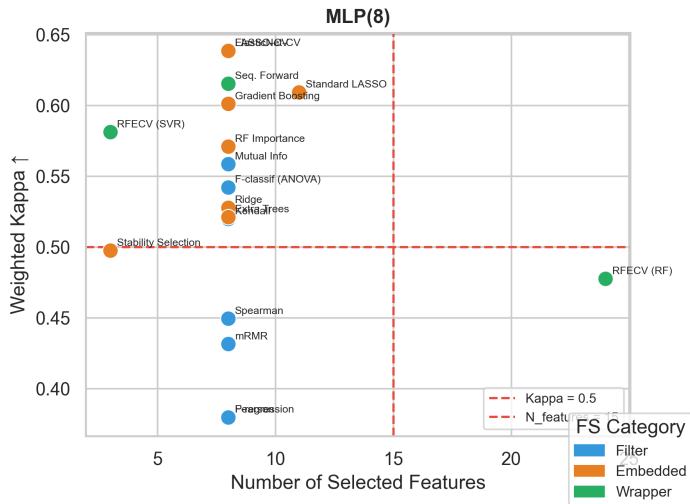
Amplitude Normalisation Fixed (min-max)



Lowest valley now = 0, highest peak = 1 (advisor's fix from Meeting #4).

Stage 2: #Features vs. Kappa Trade-off

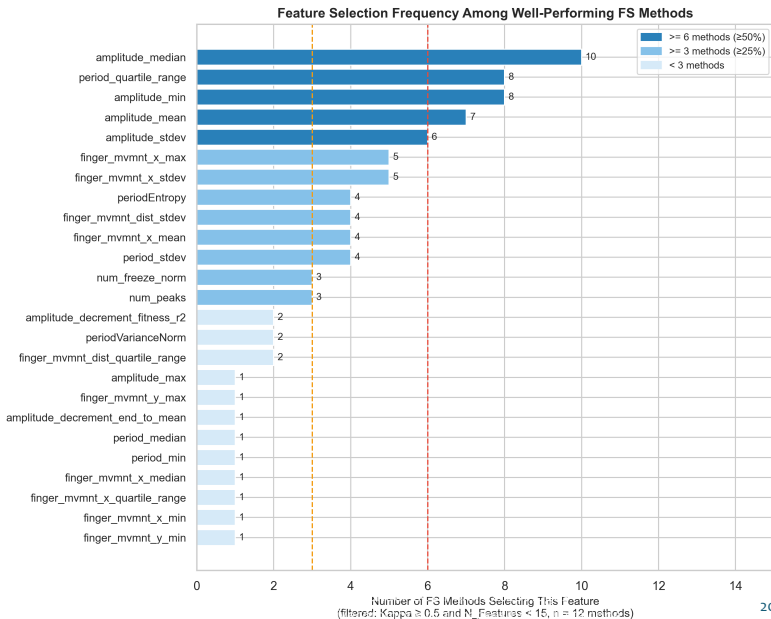
Number of Features vs Weighted Kappa
(red lines = Stage 5 filter: $\text{Kappa} \geq 0.5$ AND $\text{N_features} < 15$)



Keep FS methods with **Kappa** ≥ 0.5 and **#features** < 15 (bottom-right region)



Stage 3: Feature Ranking (Good FS Only)





Key Responses to Meeting #4

- **Amplitude normalisation:** ratio-to-max $\rightarrow$ min-max (min = 0, max = 1).
- **No artificial grouping:** every FS method sees the same scalar feature matrix.
- **Evaluation defined:** 3 predictors $\times$ LOOCV $\times$ 4 metrics — uniform across all FS methods.
- **LLM cost:** 32B LLM reserved for Stage 4 only; cheap surrogates handle Stage 2.





Open Question

How should the per-tap amplitude sequence be represented?

- (A) **Per-position** `tap_amp_norm_01..20` — 20 individual features
- (B) **Summary** `amplitude_median, amp_cv, decrement_slope, ...`

Most FS methods can only accept scalar columns — favouring (B).



Thank You



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Questions & Feedback

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